Semiconductor:

"A semiconductor is a substance which has very few free electrons at room temperature, consequently under the influence of potential difference, a semiconductor practically conducts no current".

Energy Levels and Energy Bands

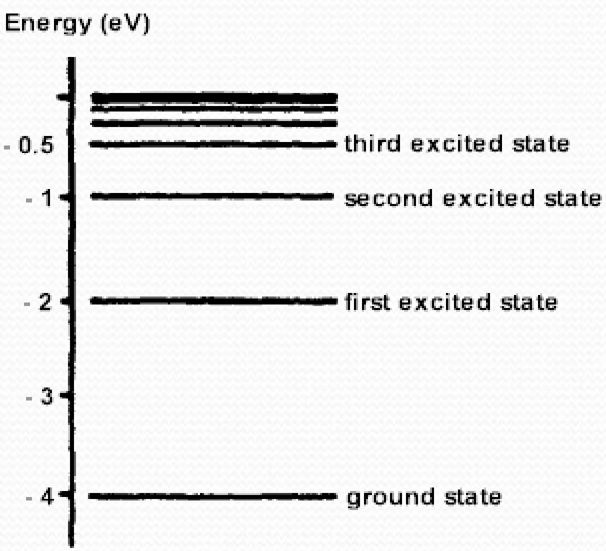
- **Energy level:**

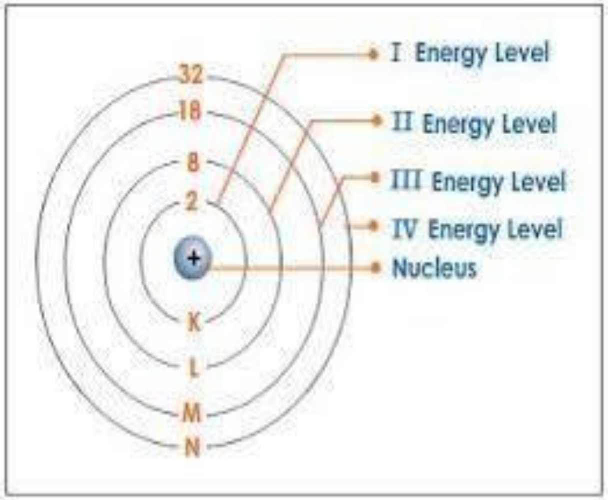
- Each orbit has fixed amount of energy associated with it.

- Electrons moving in a particular orbit possess energy of that orbit.

- The larger the orbit, the greater is its energy.

- Outer most orbit possess more energy than inner most orbit.





Energy Levels and Energy Bands

- **Energy Bands:**

"The Range of energies possessed by an electron in a solid is known as energy band"

- In single isolated atom, the electrons in any orbit possess definite energy, however an atom in a solid is greatly influenced by the closely packed neighbouring atoms, the result is that the electron in any orbit of such an atom can have a range of energies.

Important Energy Bands in Solids

- As we know that in a single isolated atom the electron in any orbit possess definite energy where as in solids they are converted to corresponding bands. There are number of energy bands in solids.

A. <u>Valance Band:</u>

“The range of energies (i-e band) possessed by valance electrons is known as valance band”.

Important Energy Bands in Solids

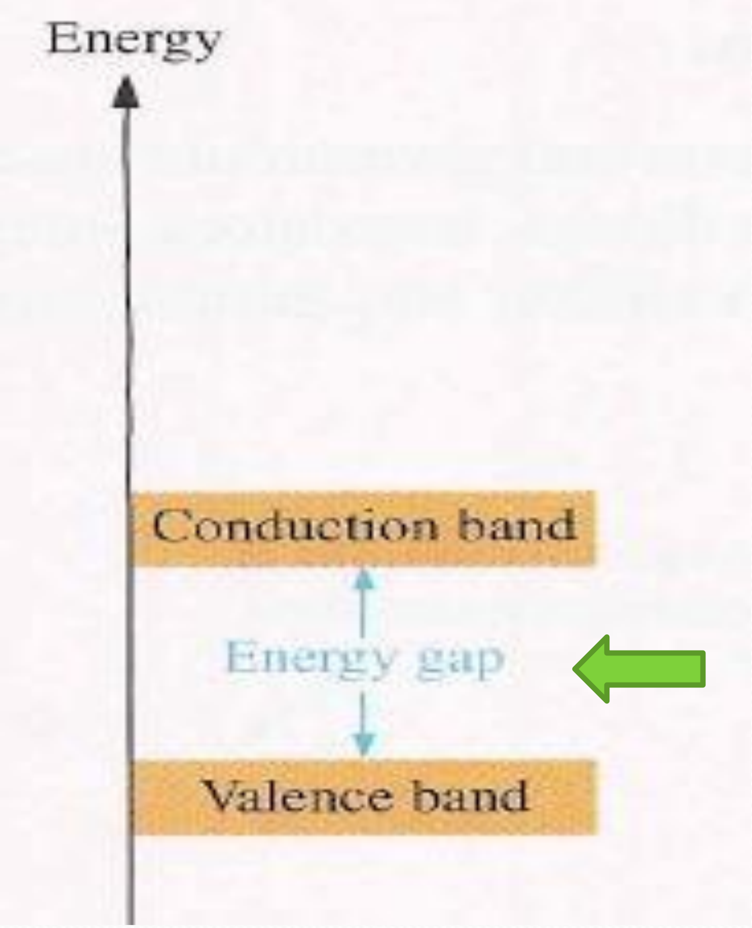
B. **Conduction Band:**

"The range of energies (i-e band) possessed by conduction electrons is known as conduction band".

C. **Forbidden Energy Gap:**

"The separation between conduction band and valance band on the energy level diagram is known as forbidden energy gap".

Important Energy Bands in Solids



Forbidden Energy Gap

Classification of Solids in Terms of Energy Bands

- Solids are classified into conductors, insulators and semiconductors.
- They are different from each other in terms of electric conductivity.
- This behaviour of solids can be easily and beautifully explained in terms of energy bands.

Classification of Solids in Terms of Energy Bands

- **<u>Insulator:</u>**

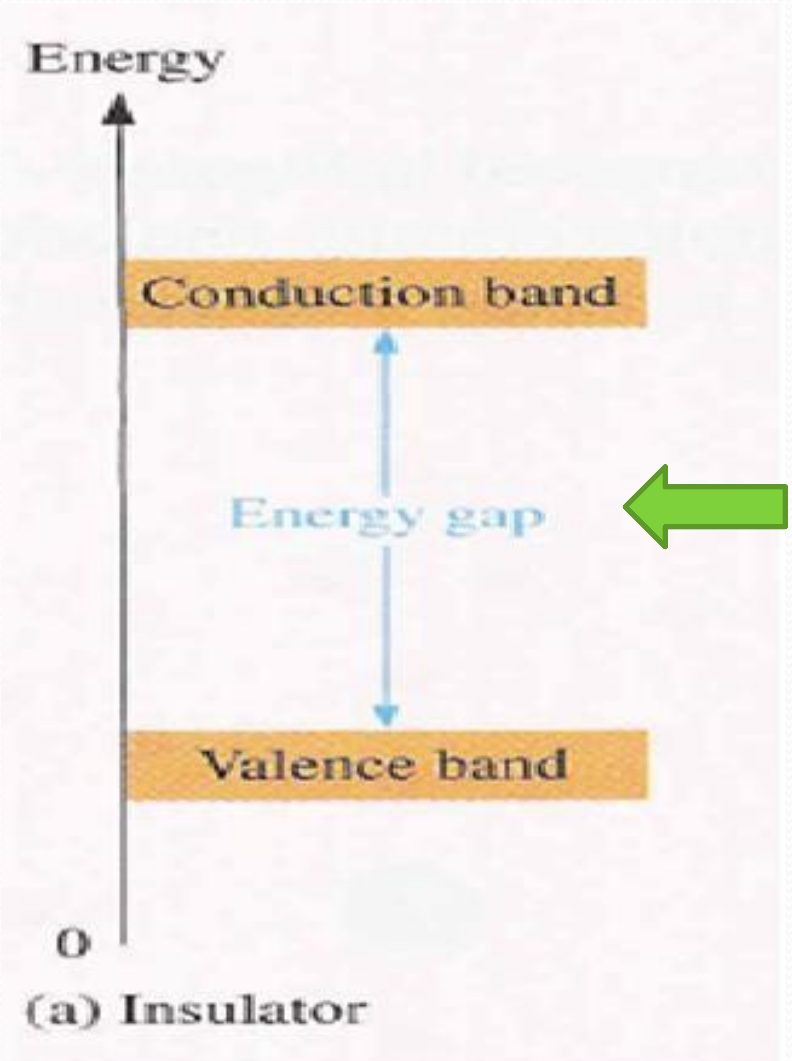
  "Insulators are those substances which do not allow passage of electric current through them".

- In terms of energy band , the valance band is full while the conduction band is empty.

- Valance band is full in insulator because electrons are tightly bonded with nucleus.

- The energy band gap between valance band and conduction band in insulators is very high i-e 15ev.

Insulators:



Energy Gap= 15ev

Classification of Solids in Terms of Energy Bands

- **Conductors:**

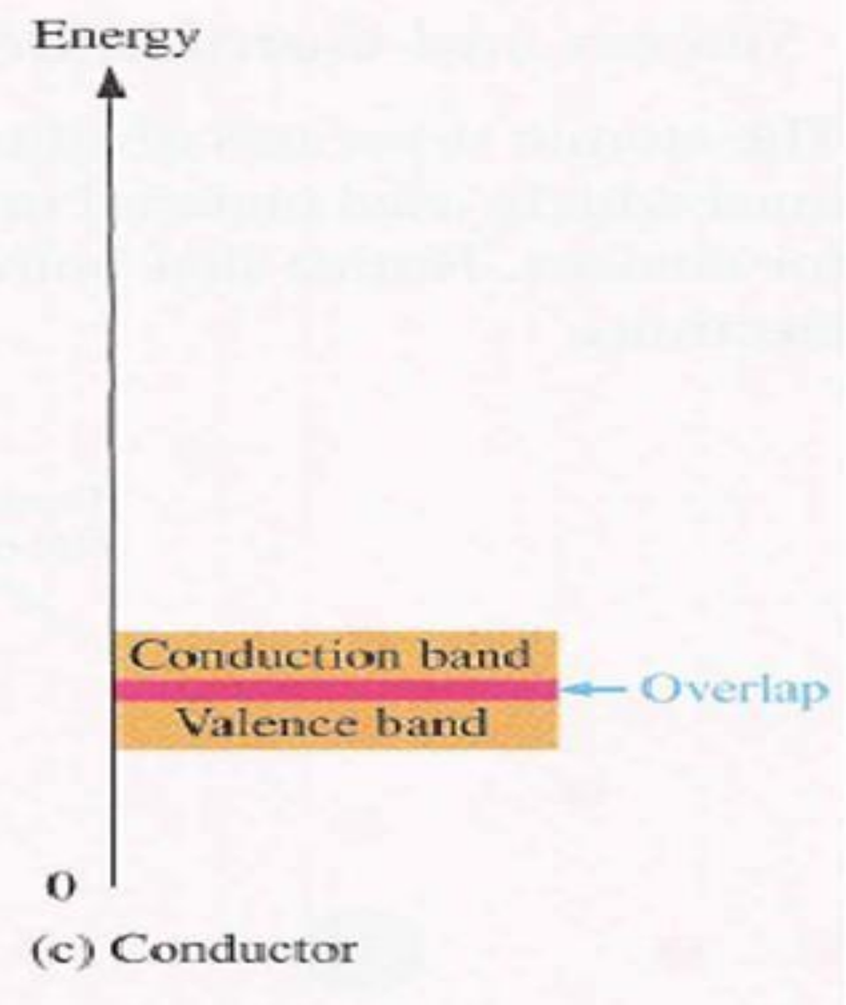
"Conductors are those substances which easily allow the passage of electric current through them".

- In terms of energy band, the valance band and conduction band overlap each other.

- In conductors, valance electrons are loosely bonded with nucleus so they become free electrons at ordinary room temperature.

- Due to the overlapping, a slightly potential difference across a conductor causes the free electrons to constitute electric current.

Conductors:



Classification of Solids in Terms of Energy Bands

- **Semiconductors:**

  "Semiconductors are those substances whose electric conductivity lie between conductor and insulators".

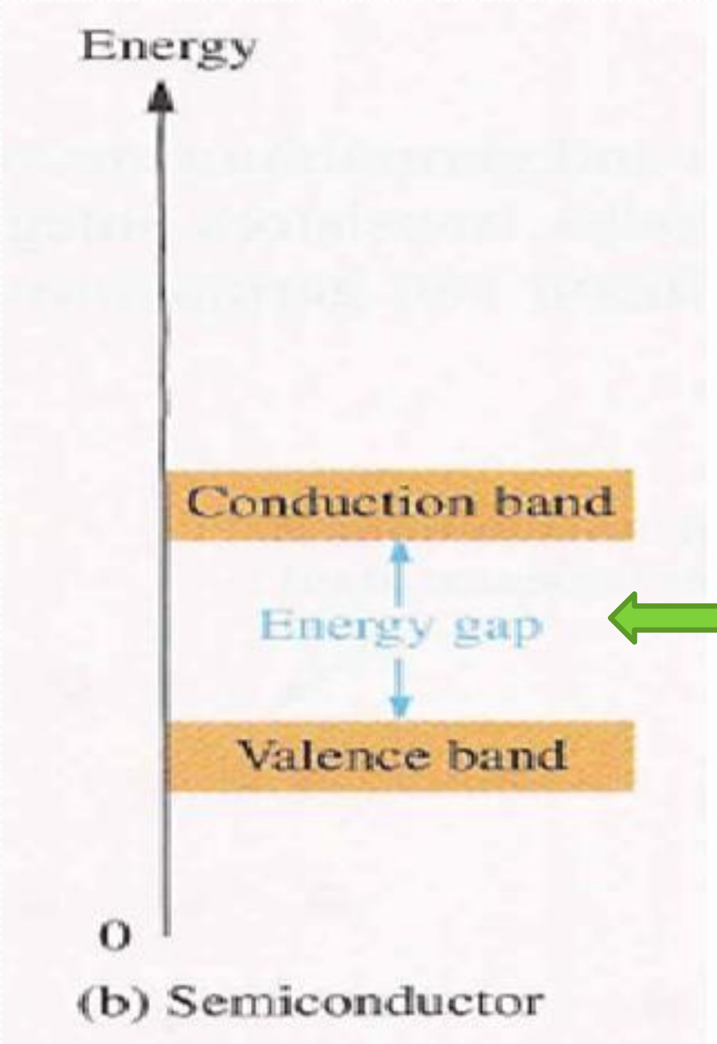
- In terms of energy band, the valance band is almost full and conduction band is almost empty.

- The energy band gap between valance and conduction band is very small i-e 1ev.

- At low temperature the valance band is completely full and conduction band is completely empty.

- At low temperature semiconductors behave like an insulator where as when temperature increases they behave like conductor.

Semiconductor:



Energy Gap = 1ev

Semiconductor

- **Properties of Semiconductor:**

- The resistivity of semiconductor is less than an insulator but more than a conductor.

- Semiconductors have negative temperature co-efficient of resistance.

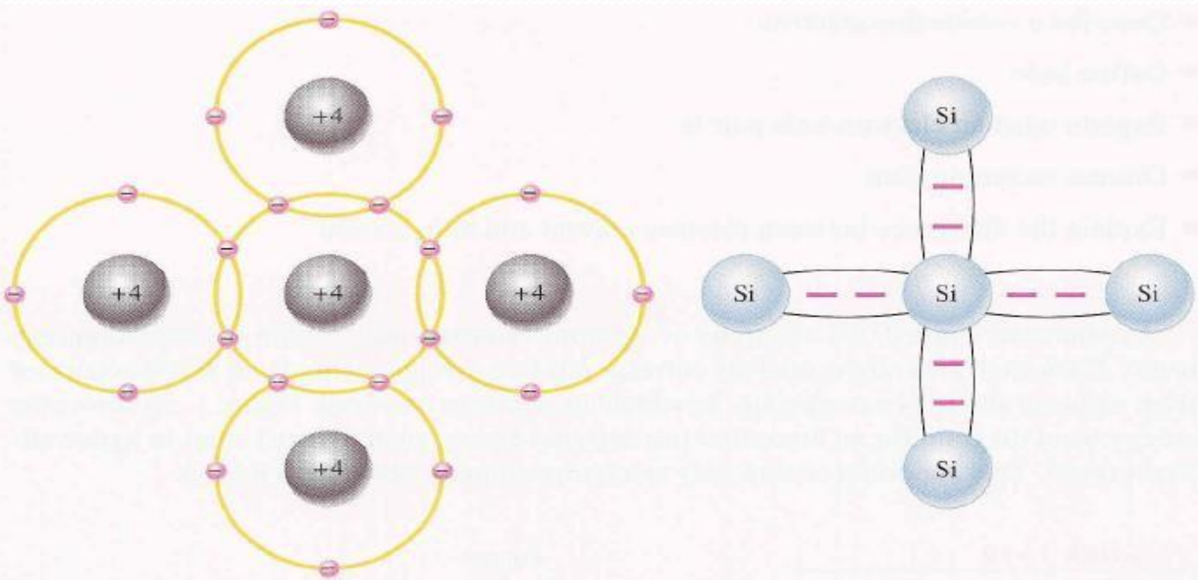
Negative temperature co-efficient of resistance??

- When a suitable metallic impurity is (e.g. Arsenic , gallium) is added to semiconductor, its current conducting properties change appreciably.

Bonding in Semiconductor

- All atoms want their valance shell completely filled or completely empty. **Why?**

  - Loosing electrons
  - Gaining electrons
  - Sharing electrons (Covalent bonding)



Commonly Used Semiconductors

- There are many semiconductors available, but very few of them have a practical application in electronics.

- The two most frequently used materials are:
  - Germanium Ge
  - Silicon Si

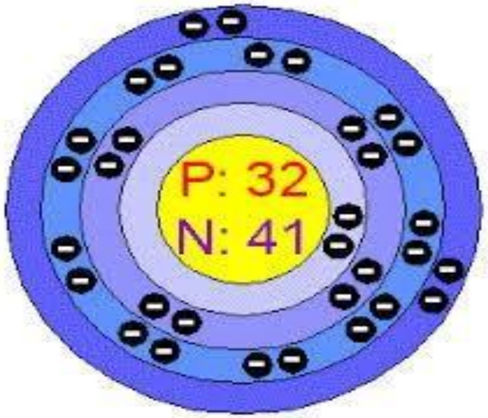
- The reason behind the frequently usage of these two materials is, the energy required to break their covalent bonds is very small.

- For Silicon it is 0.7 eV and for Germanium it is 0.3 eV

Commonly Used Semiconductors

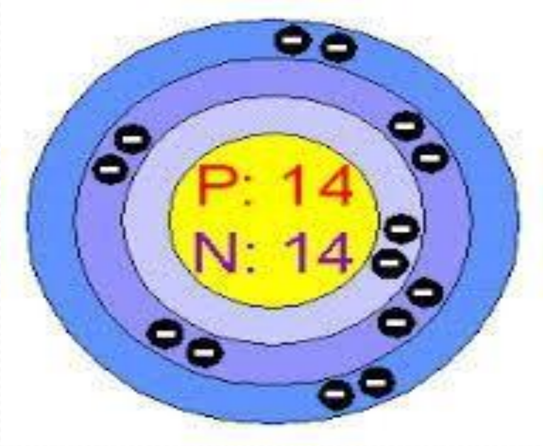
- Germanium:

  - Discovered in 1886
  - Atomic Number =32
  - Atomic Weight= 73
  - Number of Electron= 32
  - Number of Protons= 32
  - Number of Neutrons= 41
  - Valance Electrons= 4
  - Valancy= +4 or -4



Commonly Used Semiconductors

- Silicon:
  - Discovered in 1824
  - Atomic Number = 14
  - Atomic Weight= 28
  - Number of Electron= 14
  - Number of Protons= 14
  - Number of Neutrons= 14
  - Valance Electrons= 4
  - Valancy= +4 or -4



Effect of Temperature on Semiconductors

- The electrical conductivity of a semiconductor changes appreciably with temperature variation.

- <u>At Absolute Zero</u>:

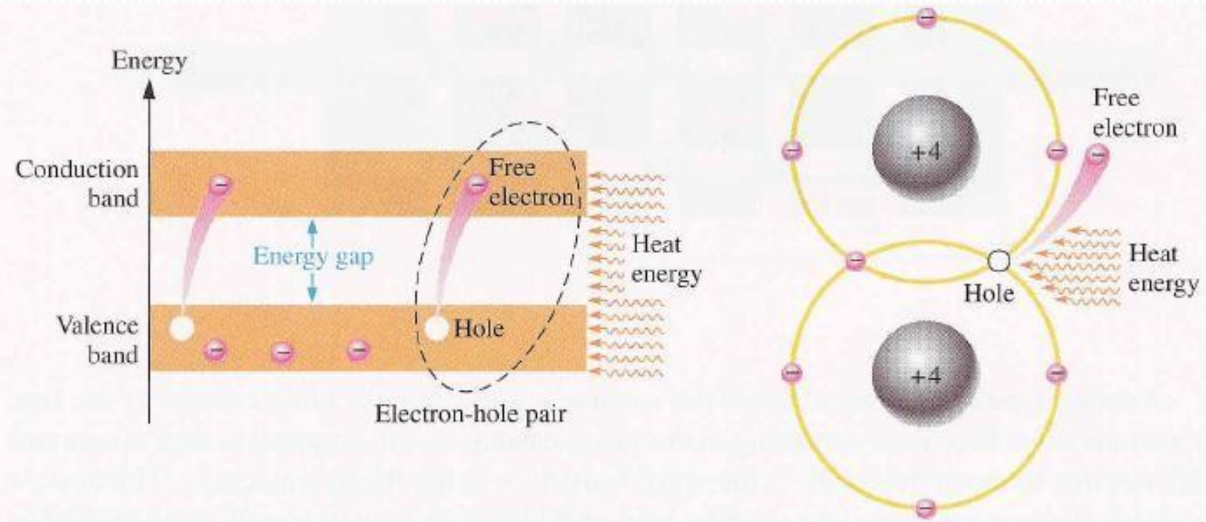
  - All electrons are tightly held by the semiconductor atom.
  - The inner orbits electrons are bound
  - The outer electrons ***i-e*** valance electrons are engaged in covalent bonding
  - The covalent bonds are very strong
  - There are no free electrons at this temperature

Effect of Temperature on Semiconductors

- Above absolute zero:
  - With increase in temperature, the covalent bonds break due to the thermal energy supplied.
  - Breaking of bonds form free electrons.
  - These free electrons can constitute a tiny current if potential difference is applied.
  - Resistance of semiconductors decrease with increase in temperature.
  - At room temperature, current through semiconductor is very small to be of any practical value.

Conduction in semiconductors

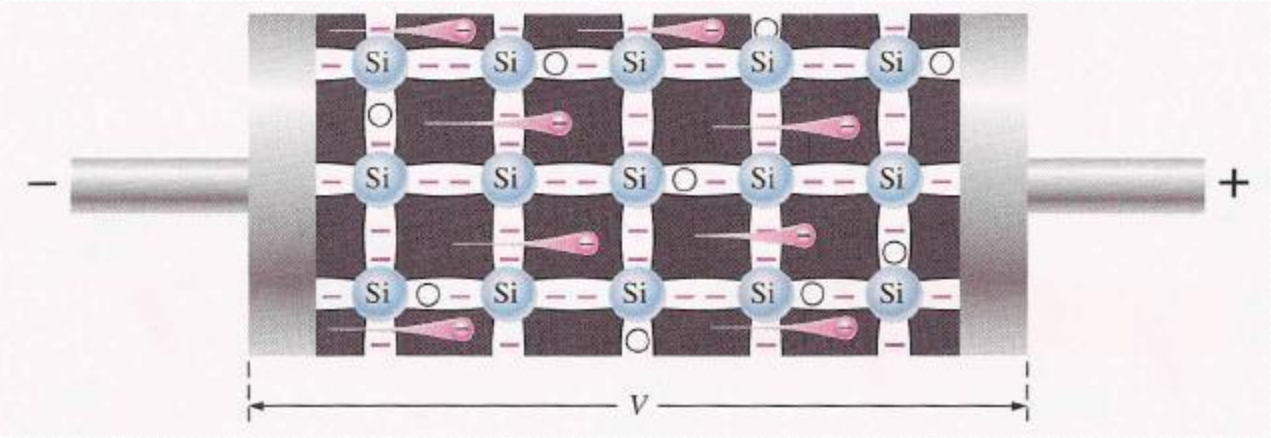
- Flow of electrons from atom to atom
- when electron jumps to conduction band (conduction electrons), a vacancy is left in valance band
  - This vacancy is called hole
- And both are called Electron-hole pair



(a) Energy diagram (b) Bonding diagram

Conduction in semiconductors

- When the voltage is applied to piece of intrinsic silicon material, thermally generated free electrons in conduction band, which are free to move are attracted towards +ve end.
- Flow of electrons from –ve end to +ve end in one direction is called electron current.



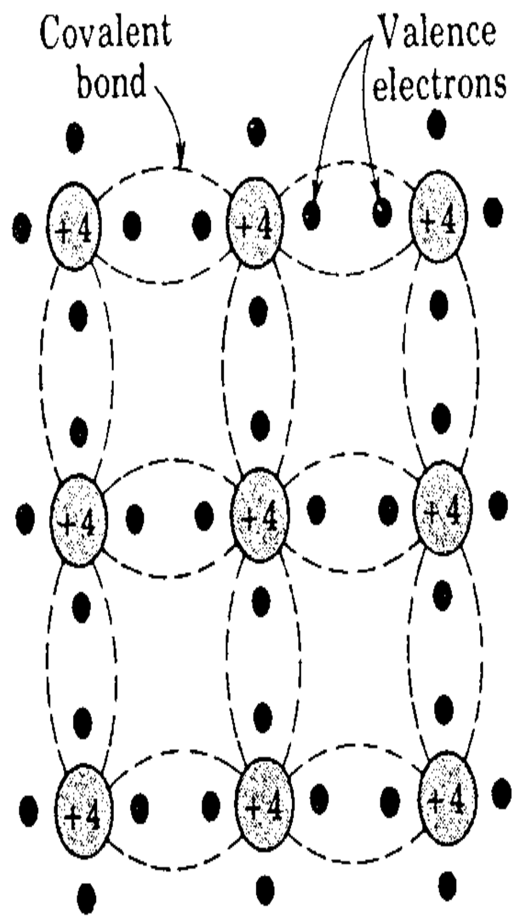
Intrinsic and Extrinsic Semiconductor

- <u>Intrinsic Semiconductor:</u>

" A Semiconductor in an extremely pure form is known as an intrinsic semiconductor"

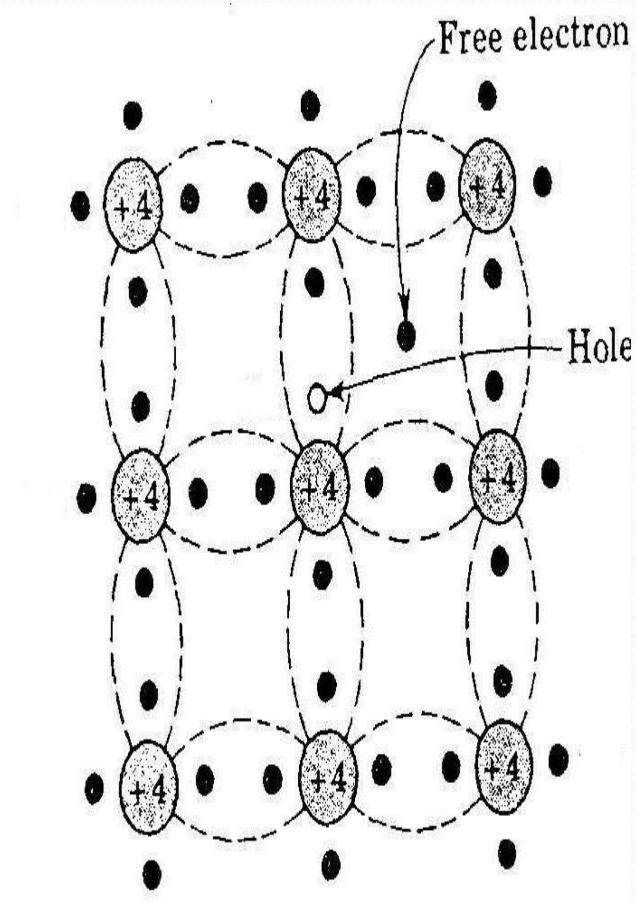
- In *intrinsic* (pure) silicon, atoms join together by forming *covalent bonds*.

- Each atom shares its valence electrons with each of four adjacent neighbours effectively filling its outer shell.



Intrinsic Semiconductor

- At room temperature some of the electrons are able to acquire sufficient thermal energy to break freely from their bond.
- Whenever an electron leaves its position, it leaves a vacancy known as a *hole*.
- The process is known as *electron-hole pair generation*



Intrinsic Semiconductor

- A free electron can move through the body of the material until it encounters another broken bond where it is drawn in to complete the bond or *recombines*.

- When an voltage is applied, the holes move in one direction and the electrons in the other.

- However both current components are in the direction of the field.

- The conduction is Ohmic, i.e. current is proportional to the applied voltage (field)

Extrinsic Semiconductor

- "When a small amount of suitable impurity is added in pure semiconductor to increase its conduction properties, the semiconductor is known as Extrinsic Semiconductor".

- The Process of adding impurities to a semiconductor is known as ***doping.***

- The amount and type of such impurities have to be closely controlled during the preparation of extrinsic semiconductor.

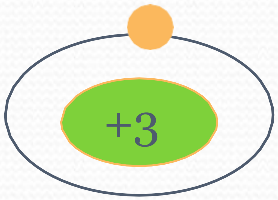
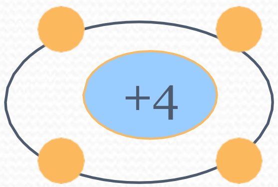
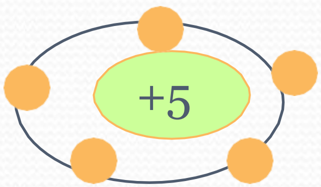
- Generally for **10^6** atoms of semiconductor, **one** atom of impurity is added

Extrinsic Semiconductor

- If pentavalent impurity (having 5 valance electrons) is added to the semiconductor, a large number of free electrons are produced in the semiconductor.
- When trivalent impurity (having 3 valance electrons) is added to the semiconductor, a large number of holes are produced in the semiconductor.
- Depending upon the type of impurity added, extrinsic semiconductor are classified into:

1. N-type Semiconductor
2. P-type Semiconductor

Semiconductor Elements in Periodic Table

Group III	Group IV	Group V
		
Boron (B)	Carbon (C)	Nitrogen (N)
Aluminium (Al)	Silicon (Si)	Phosphorus (P)
Gallium (Ga)	Germanium (Ge)	Arsenic (As)
Indium (In)	Tin (Sn)	Antimony (Sb)

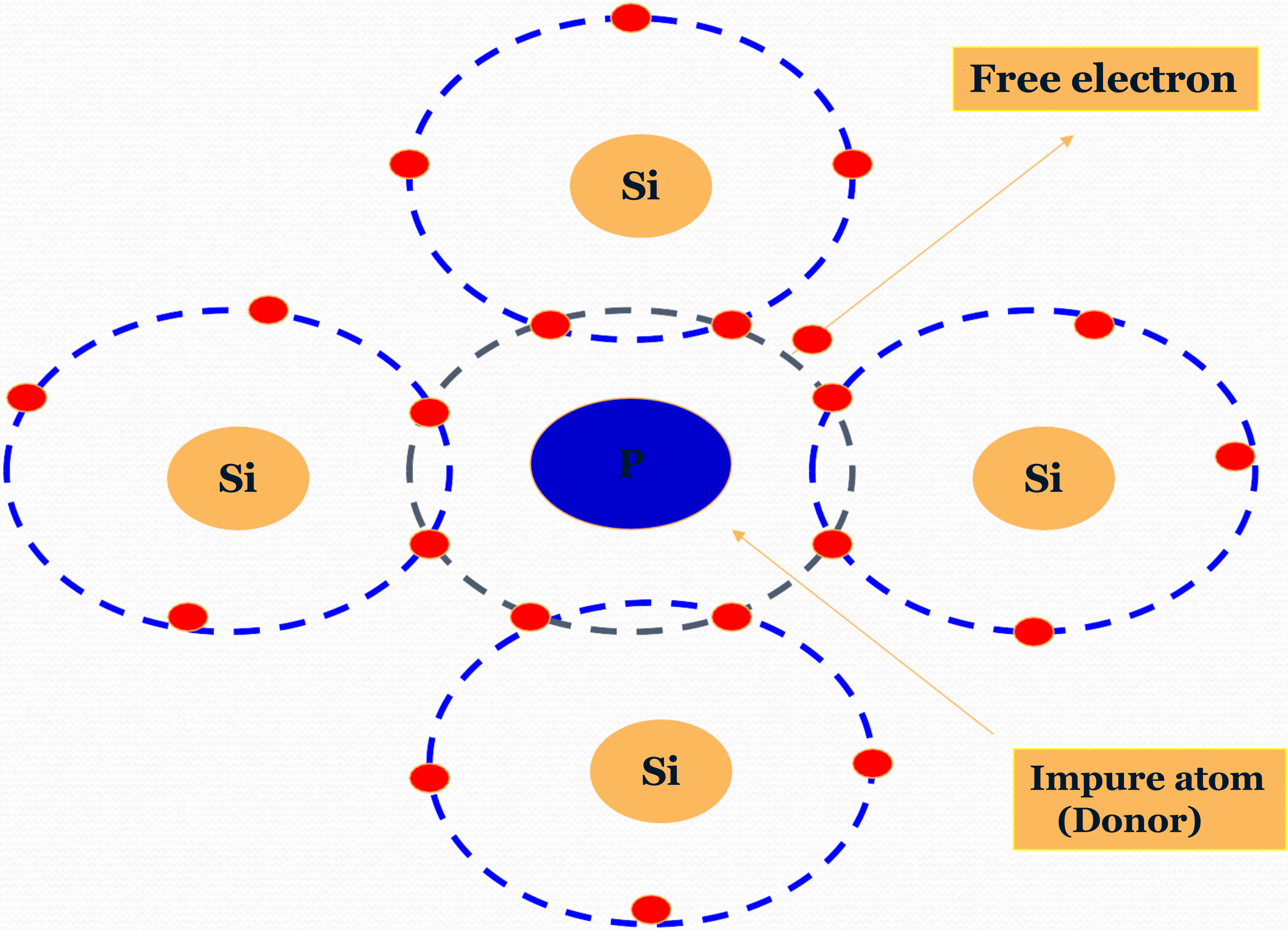
N-type Semiconductor

- When any pentavalent element such as **Phosphorous**, **Arsenic** or **Antimony** is added to the intrinsic Semiconductor , four electrons are involved in covalent bonding with four neighboring pure Semiconductor atoms.

- The fifth electron is weakly bound to the parent atom. And even for lesser thermal energy it is released Leaving the parent atom positively ionized.

- The intrinsic Semiconductors doped with pentavalent impurities are called N-type Semiconductors.

N-type Semiconductor



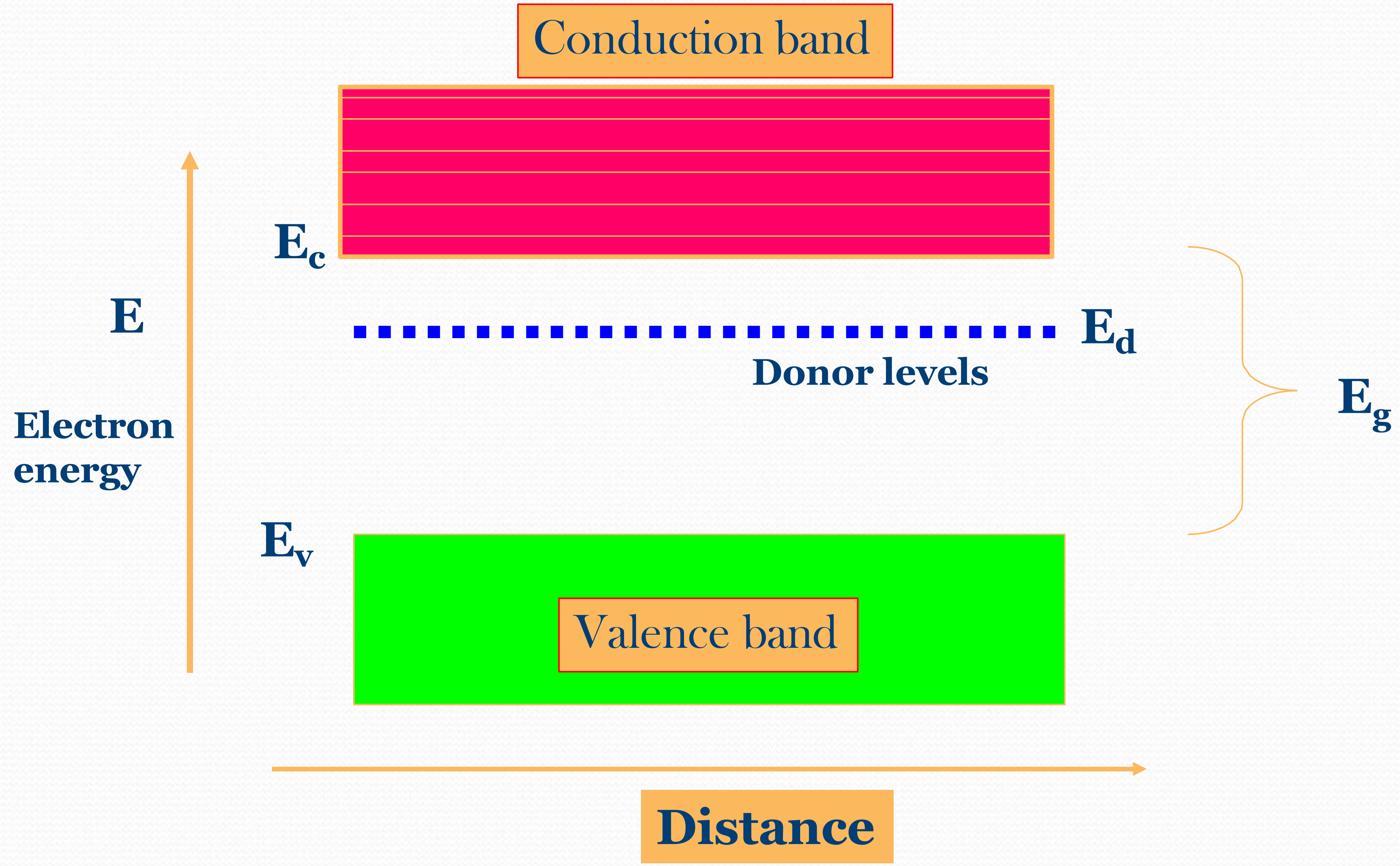
N-type Semiconductor

- The energy level of fifth electron is called donor level.

- The donor level is close to the bottom of the conduction band most of the donor level electrons are excited in to the conduction band at room temperature and become the **majority charge carriers**.

- Hence in N-type Semiconductors electrons are **majority carriers** and holes are **minority carriers**.

N-type Semiconductor

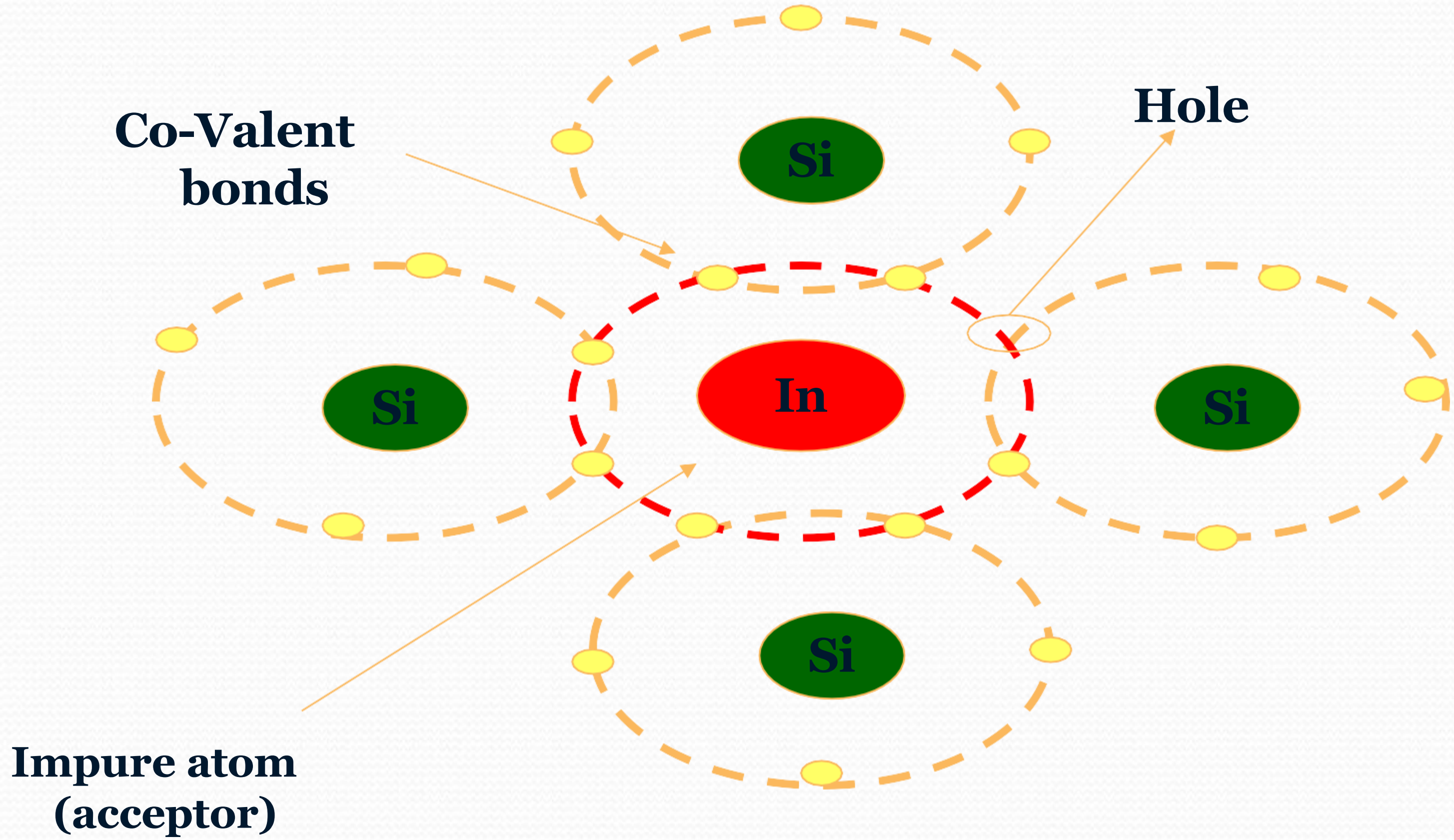


P-type Semiconductor

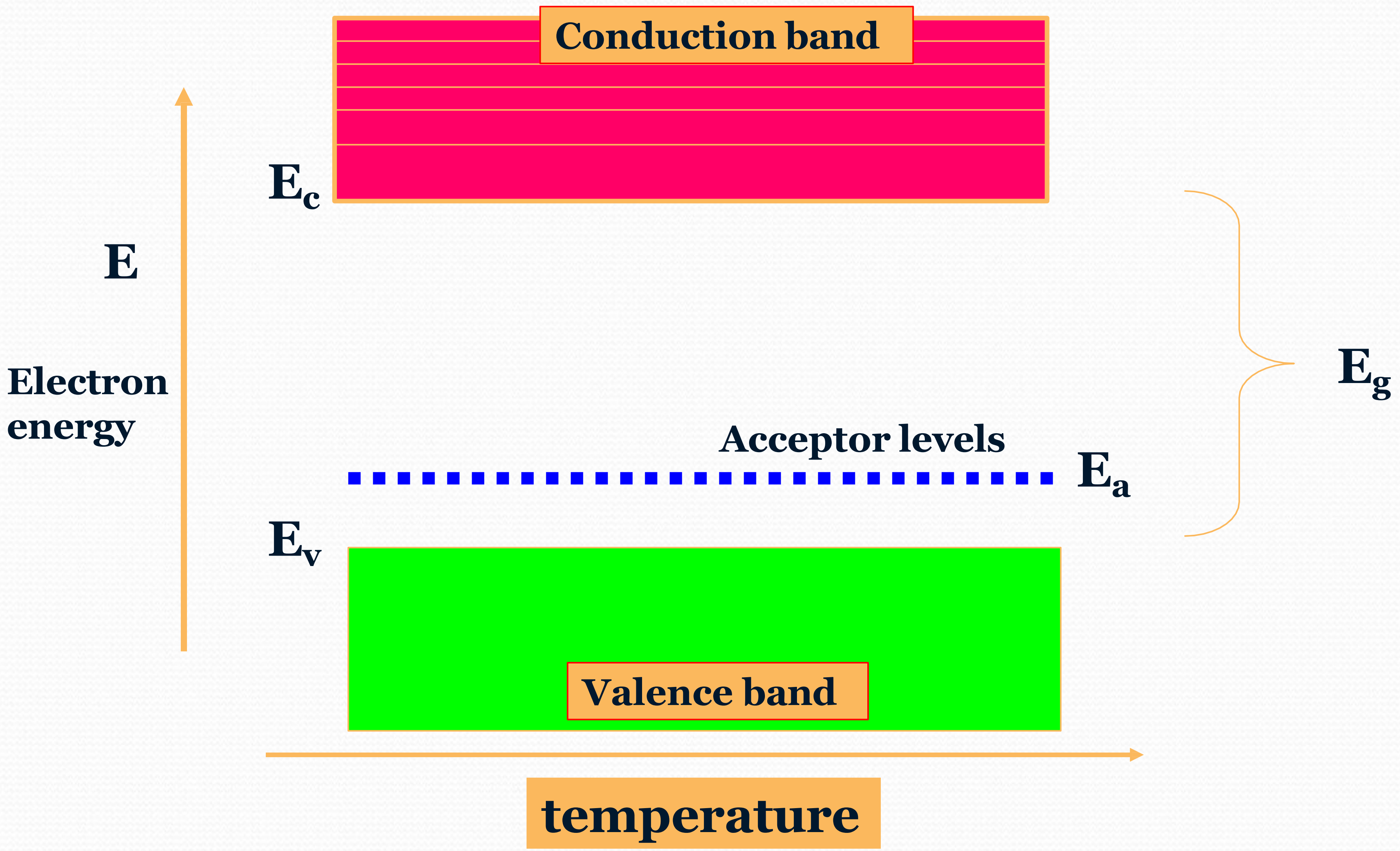
- When a trivalent elements such as **Al, Ga or Indium** have three electrons in their outer most orbits , added to the intrinsic semiconductor all the three electrons of Indium are engaged in covalent bonding with the three neighboring Si atoms.
- Indium needs one more electron to complete its bond.
- This electron maybe supplied by Silicon , there by creating a vacant electron site or hole on the semiconductor atom.
- Indium accepts one extra electron, the energy level of this impurity atom is called **acceptor level** and this acceptor level lies just above the valence band.

P-type Semiconductor

- These type of trivalent impurities are called **acceptor impurities** and the semiconductors doped the acceptor impurities are called **P-type semiconductors**.



P-type Semiconductor



N-type and P-type Semiconductor

Summary

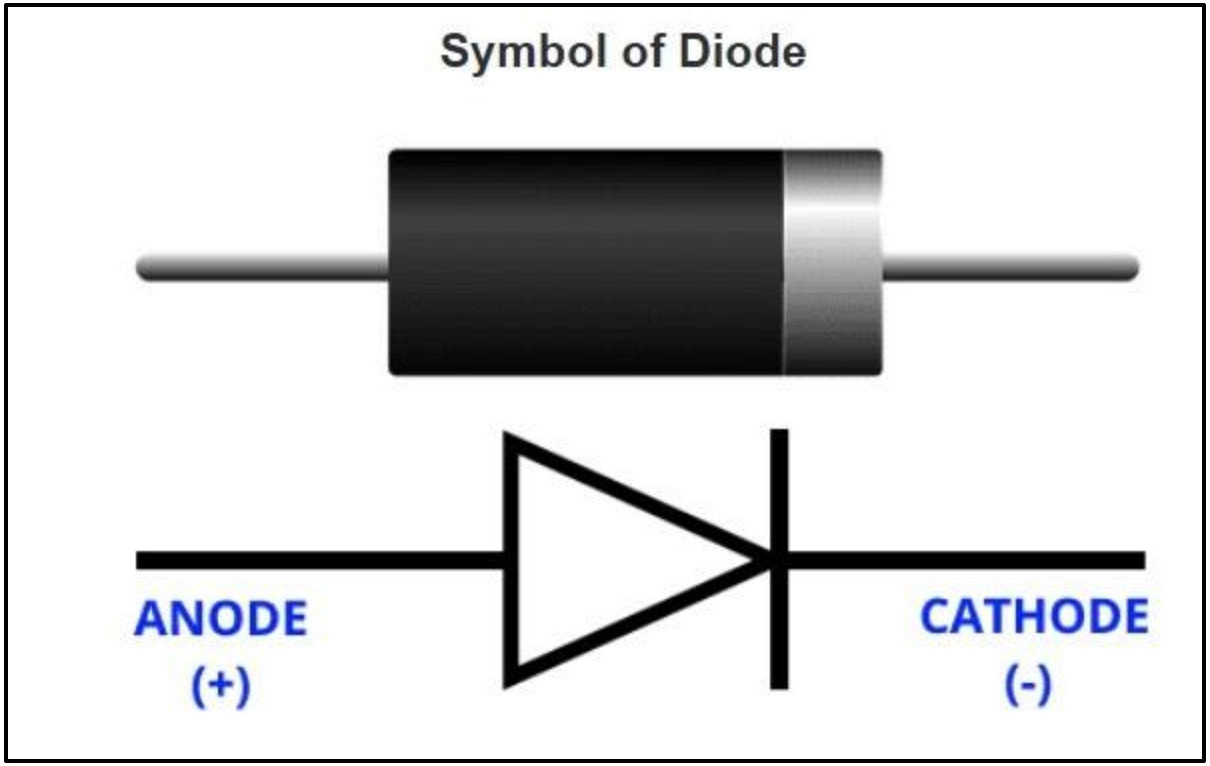
N-type Semiconductor	P-type semiconductor
•In an N-type semiconductor, conduction is mainly due to electrons (negative charges) •Positive charges (holes) are the minority carriers.	•In a P-type semiconductor, conduction is mainly due to holes (positive charges) •Negative charges (electrons) are the minority carriers.

Diode

Diode

- ***Di "= Two****, and "****Ode "= Electrodes*** *i.e a device or component having two electrodes Anode "+" (P) and Cathode "-" (N).*
- A diode is a two-terminal unidirectional power electronics device. The semiconductor diode is the first invention in a family of semiconductor electronics devices. After that many types of diodes are invented. But today also the most commonly used diode is a semiconductor diode.
- Generally, silicon is used to make a diode. But another semiconductor material like germanium or germanium arsenide is also used.
- A diode allows current to flow only in one direction and it blocks the current in another direction. It offers low resistance (ideally zero) in one direction and it offers a high resistance (ideally infinite) in another direction.

Cont.



Construction of Diode

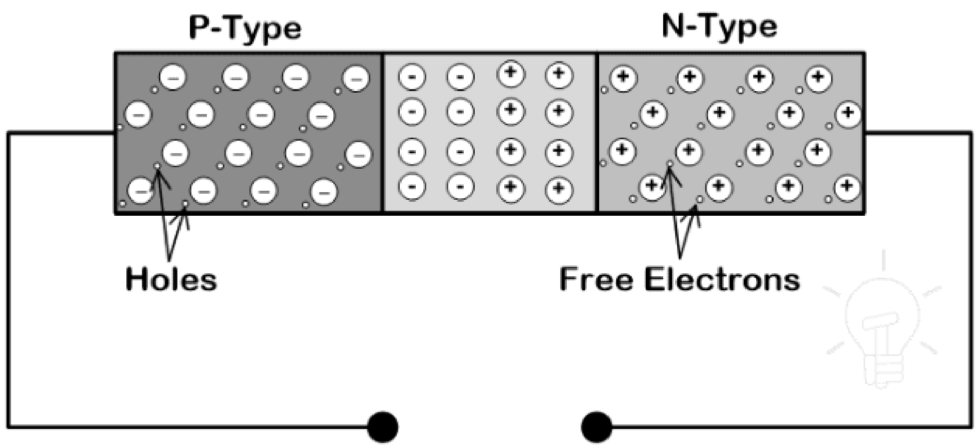
- There are two types of semiconductor material; Intrinsic and Extrinsic semiconductor. An intrinsic semiconductor is a pure semiconductor in which hole and electrons are available in equal numbers at room temperature. In an extrinsic semiconductor, impurities are added to increase the number of holes or the number of electrons. These impurities are tri-valent (boron, indium, aluminum) or pentavalent (phosphorous, Arsenic, Antimony).
- A semiconductor diode has two layers. One layer is made of a P-type semiconductor layer and the second layer is made of an N-type semiconductor layer.
- If we add trivalent impurities in silicon or germanium, a greater number of holes are present and it is a positive charge. Hence, this layer is known as the P-type layer.
- If we add pentavalent impurities in silicon or germanium, a greater number of electrons are present and it is a negative change. Hence, this layer is known as the N-type layer.

Cont.

- The diode is formed by joining both N-type and P-type semiconductors together. This device is a combination of P-type and N-type semiconductor material hence it is **also known as PN Junction Diode**.
- A junction is formed between the P-type and N-type layers. This junction is known as PN junction.
- A diode has two terminals; one terminal is taken from the P-type layer and it is known as Anode. The second terminal is taken from the N-type material and it is known as Cathode.

Working of Diode

In the N-type region, electrons are the majority charge carriers and holes are minority charge carriers. In the P-type region, the holes are majority charge carrier and the electrons are negative charge carriers. Because of the concentration difference, majority charge carriers diffuse and recombine with the opposite charge. It makes a positive or negative ion. These ions are collected at the junction. And this region is known as the depletion region.



When anode terminal of diode is connected with a negative terminal and cathode is connected with the positive terminal of a battery, the diode is said to be connected in reverse bias.

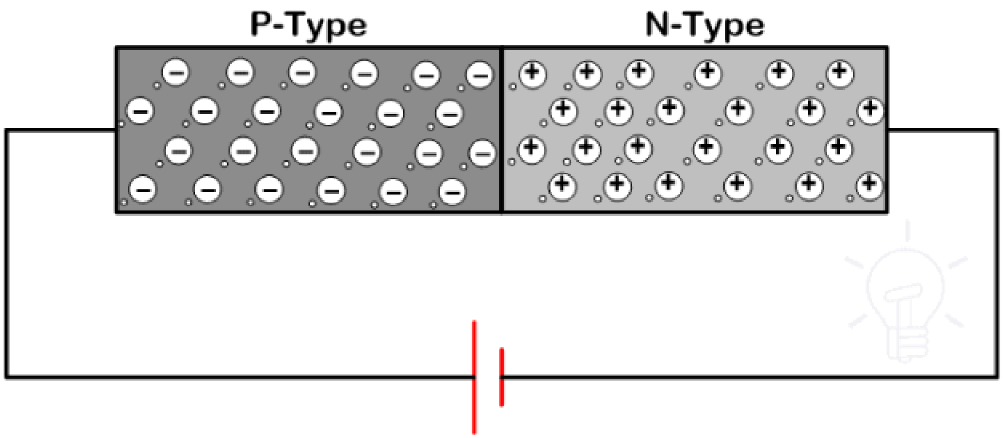
Similarly, when anode terminal is connected with a positive terminal and cathode is connected with the negative terminal of the battery, the diode is said to be connected in forward bias.

Operation of Diode in Forward Bias Condition

- When the anode is connected with the positive terminal of the battery and cathode is connected with a negative terminal of the battery, the anode is positive with respect to the cathode. And the diode is said to be connected in forward bias
- Now, we gradually increase the supply voltage. If we increase a small voltage, the majority charge carrier does not get sufficient energy to cross the depletion region.
- In the forward bias conditions, the width of the depletion region is very small. If we increase the voltage more than forward breakover voltage, the majority charge carrier gets enough energy to cross the depletion region.
- For silicon forward breakover voltage is 0.7V and for germanium forward breakover voltage is 0.3V.
- When supply voltage increases more than this voltage, the majority charge carriers flowing through the circuit and it made the diode conducting.
- In this mode of operation, a very small amount of drop occurs. This drop is known as an on-state voltage drop.

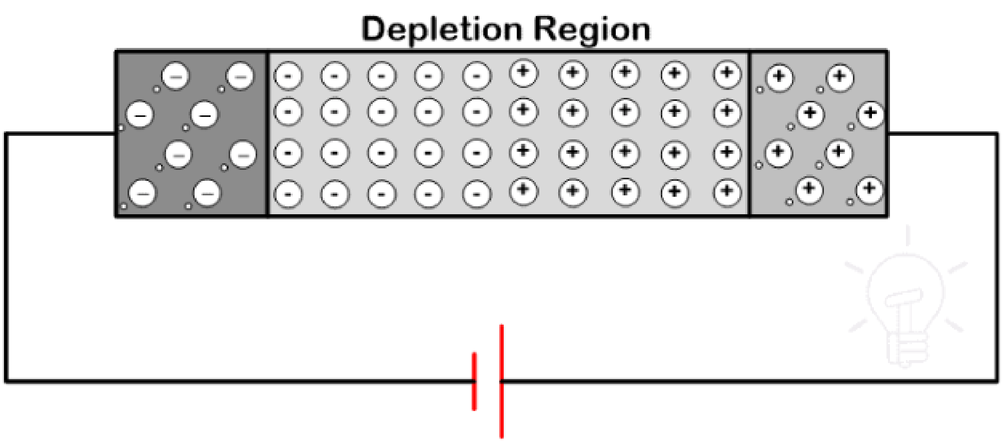
Cont.

The connection diagram of this mode is as shown in the figure.



Operation of Diode in Reverse Bias Condition

- The diode is connected in reverse bias. In this condition, free electrons diffusing into the P-type regions and recombine with holes. It will create negative ions. Similarly, holes diffuse into the N-type region and recombine with electrons. It will create positive ions.
- The connection diagram is as shown in the below figure.



Cont.

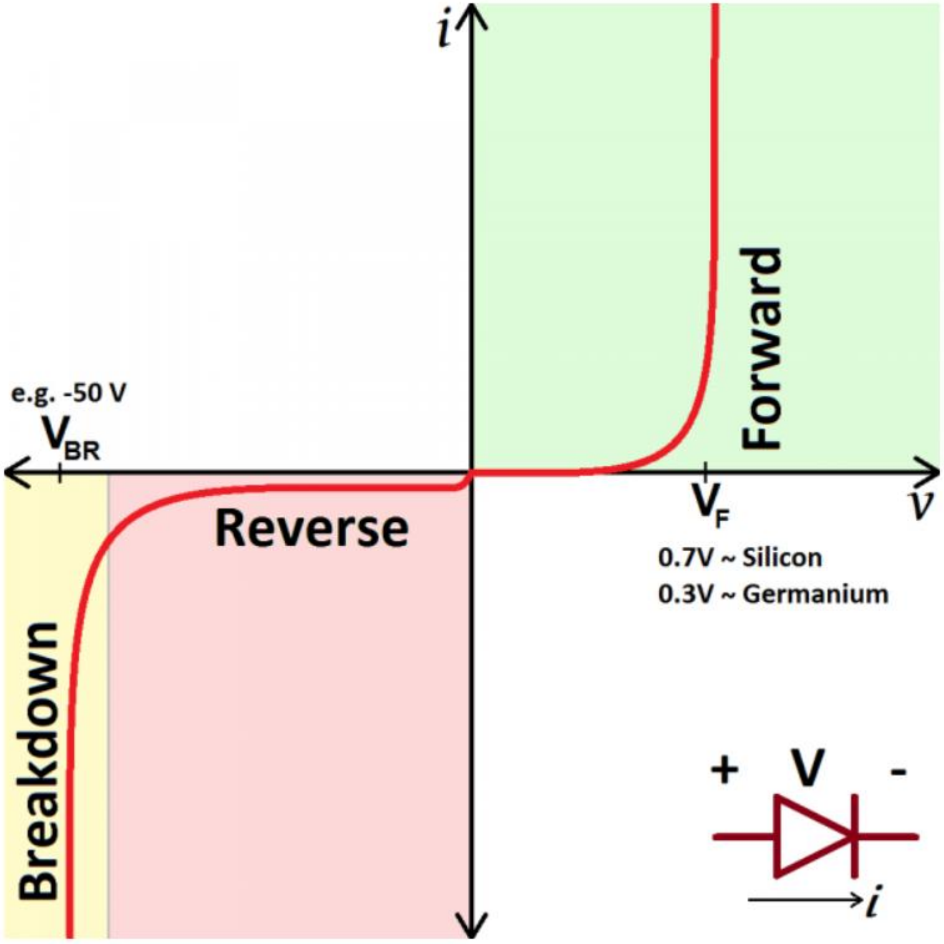
- When such voltage is applied to the circuit, immobile ions create a depletion region as shown in the previous figure. The width of the depletion region is large. Hence, no more hole or electron crosses the junction.
- It cannot create a flow of electrons or holes even if it is supplied at rated voltage. Hence, it is not possible to flow the current through the diode and it behaves as an open switch.
- Here a very small amount of current will flow through the circuit. This current is known as reverse saturation current or reverses leakage current. This current flow due to the minority charge carriers. This current is not high enough to conduct the diode.
- If we increase the voltage up to reverse breakover voltage, the minority charge carriers get high kinetic energy and collide with atoms. In this condition, the number of covalent bonds broken and a huge number of electrons-holes pair generates a huge amount of flow of current.
- Due to this high amount of current, a diode may get damaged. Hence, in general condition, the diode is not connected in reverse bias.

Voltage-Current Characteristics of Diode

- *Ideally*, diodes will block any and all current flowing the reverse direction, or just act like a short-circuit if current flow is forward. Unfortunately, actual diode behavior isn't quite ideal. Diodes do consume some amount of power when conducting forward current, and they won't block out all reverse current. Real-world diodes are a bit more complicated, and they all have unique characteristics which define how they actually operate.

- The most important diode characteristic is its voltage-current (*V*-*I*) relationship. This defines what the current running through a component is, given what voltage is measured across it. Resistors, for example, have a simple, linear *v*-*i* relationship (Ohm's Law). The *V*-*I* curve of a diode, though, is entirely *non*-linear. It looks something like this as shown in next slide.

V-I Characteristic Curve



Cont.

Depending on the voltage applied across it, a diode will operate in one of three regions:

- **Forward bias**: When the voltage across the diode is positive the diode is "on" and current can run through. The voltage should be greater than the forward voltage (V_F) in order for the current to be anything significant.
- **Reverse bias**: This is the "off" mode of the diode, where the voltage is less than V_F but greater than $-V_{BR}$. In this mode current flow is (mostly) blocked, and the diode is off. A *very* small amount of current (on the order of nA) -- called reverse saturation current -- is able to flow in reverse through the diode.
- **Breakdown**: When the voltage applied across the diode is very large and negative, lots of current will be able to flow in the reverse direction, from cathode to anode.

Forward Voltage

- In order to "turn on" and conduct current in the forward direction, a diode requires a certain amount of positive voltage to be applied across it. The typical voltage required to turn the diode on is called the *forward voltage* (V_F). It might also be called either the *cut-in voltage* or *on-voltage*.
- As we know from the *i-v* curve, the current through and voltage across a diode are interdependent. More current means more voltage, less voltage means less current. Once the voltage gets to about the forward voltage rating, though, large increases in current should still only mean a very small increase in voltage. If a diode is fully conducting, it can usually be assumed that the voltage across it is the forward voltage rating.

Cont.

- A specific diode's V_F depends on what semiconductor material it's made out of. Typically, a silicon diode will have a V_F around **0.6-1V**. A germanium-based diode might be lower, around 0.3V. The *type* of diode also has some importance in defining the forward voltage drop; light-emitting diodes can have a much larger V_F, while Schottky diodes are designed specifically to have a much lower-than-usual forward voltage.

Breakdown Voltage

- If a large enough negative voltage is applied to the diode, it will give in and allow current to flow in the reverse direction. This large negative voltage is called the breakdown voltage. Some diodes are actually designed to operate in the breakdown region, but for most normal diodes it's not very healthy for them to be subjected to large negative voltages.
- For normal diodes this breakdown voltage is around -50V to -100V, or even more negative.

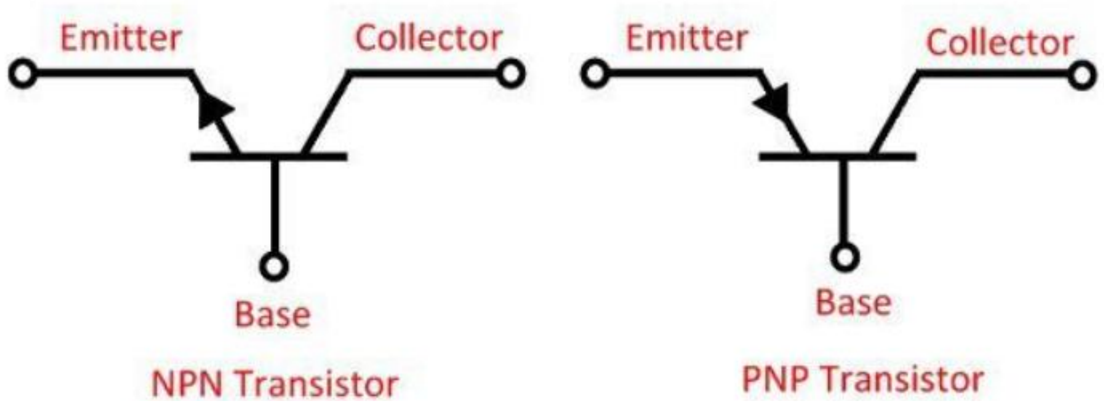
Transistor

Transistor

- The transistor is a semiconductor device which transfers a weak signal from low resistance circuit to high resistance circuit.
- The words **trans** mean **transfer** and **istor** mean **resistance property offered to the junctions.**
- In other words, it is a switching device which regulates and amplify the electrical signal likes voltage or current.
- The transistor consists two PN diode connected back to back. It has three terminals namely emitter, base and collector. The base is the middle section which is made up of thin layers. The right part of the diode is called emitter diode and the left part is called collector-base diode.

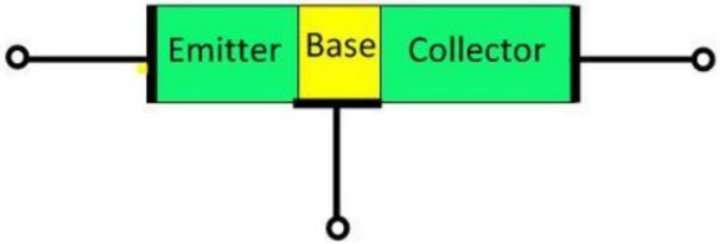
Transistor Symbols

There are two types of transistor, namely NPN transistor and PNP transistor. The transistor which has two blocks of N-type semiconductor material and one block of P-type semiconductor material is known as NPN transistor. Similarly, if the material has one layer of N-type material and two layers of P-type material then it is called PNP transistor. The symbol of NPN and PNP is shown



Transistor Terminals

The transistor has three terminals namely, emitter, collector and base. The terminals of the diode are explained below in details



Emitter: The section that supplies the large section of majority charge carrier is called emitter. The emitter is always connected in forward biased with respect to the base so that it supplies the majority charge carrier to the base. The emitter-base junction injects a large amount of majority charge carrier into the base because it is heavily doped and moderate in size.

Cont.

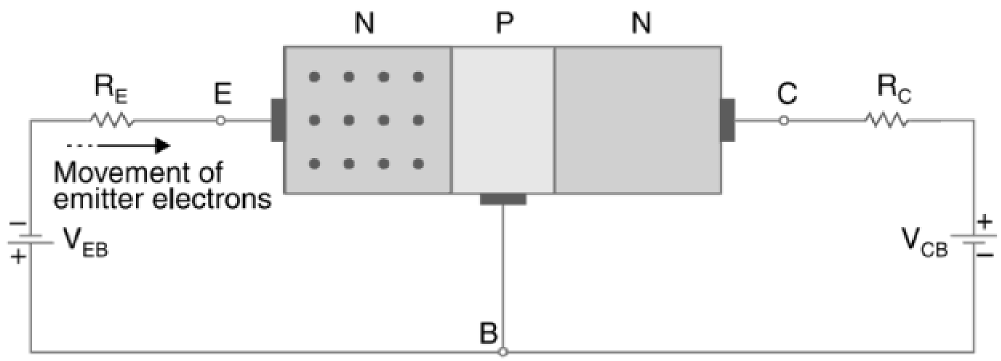
- **Collector:** The section which collects the major portion of the majority charge carrier supplied by the emitter is called a collector. The collector-base junction is always in reverse bias. Its main function is to remove the majority charges from its junction with the base. The collector section of the transistor is moderately doped, but larger in size so that it can collect most of the charge carrier supplied by the emitter.
- **Base :** The middle section of the transistor is known as the base. The base forms two circuits, the input circuit with the emitter and the output circuit with the collector. The emitter-base circuit is in forward biased and offered the low resistance to the circuit. The collector-base junction is in reverse bias and offers the higher resistance to the circuit. The base of the transistor is lightly doped and very thin due to which it offers the majority charge carrier to the base.

NPN Transistor

NPN transistor consists of two N-type semiconductor regions separated by a thin P-type semiconductor region. The three regions of the transistor are called emitter, base, and collector. In an NPN transistor, the emitter-base junction is forward biased while the collector-base junction is reverse biased. Electrons are the majority charge carriers in an NPN transistor, while holes are the minority carriers

Working of NPN

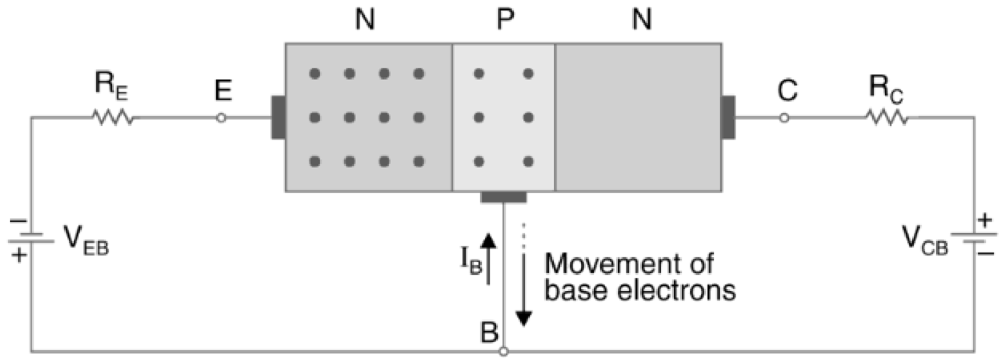
- The emitter-base junction (E–B junction) is forward biased by the supply voltage V_{BE}. In forward bias, the N-type emitter is connected to the negative terminal and the P-type base is connected to the positive terminal of the battery.
- Since the emitter is heavily doped, it contains a large number of free electrons. Due to the applied voltage, these electrons start moving from the emitter toward the base region.
- It may be noted that the direction of conventional current is opposite to the flow of electrons.



(*a*) Electrons in the emitter region.

Cont.

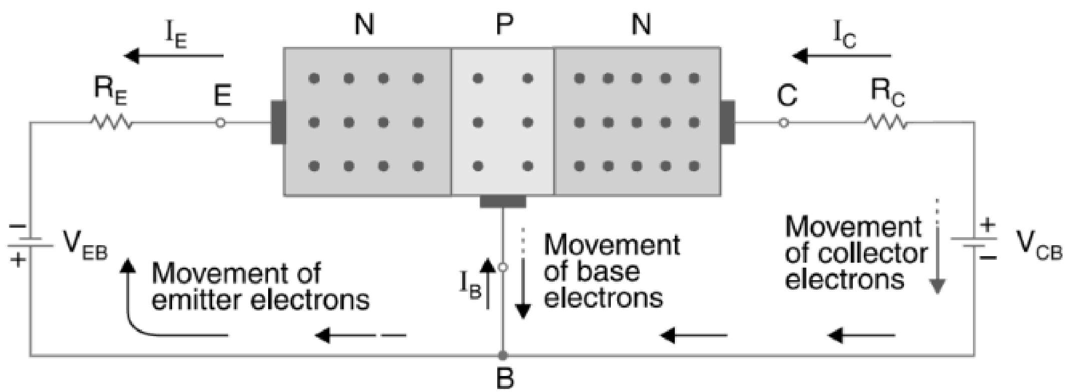
- The electrons entering the base region recombine with a few holes present in the thin and lightly doped base. This recombination produces a small base current I_B.
- Because the base region is very thin, only a small number of electrons recombine. Most electrons continue moving toward the collector region.



(*b*) Electrons in the base region.

Cont.

- The collector-base junction (C–B junction) is reverse biased by the supply voltage V_{CB} . In reverse bias, the N-type collector is connected to the positive terminal and the P-type base is connected to the negative terminal of the battery.
- Due to this reverse bias, the collector attracts the remaining electrons coming from the base region. These electrons enter the collector and produce collector current I_C .



(c) Electrons entered in the collector region.

Cont.

- It may be noted that the direction of conventional current is opposite to the flow of electrons
- Thus, most electrons supplied by the emitter are collected by the collector

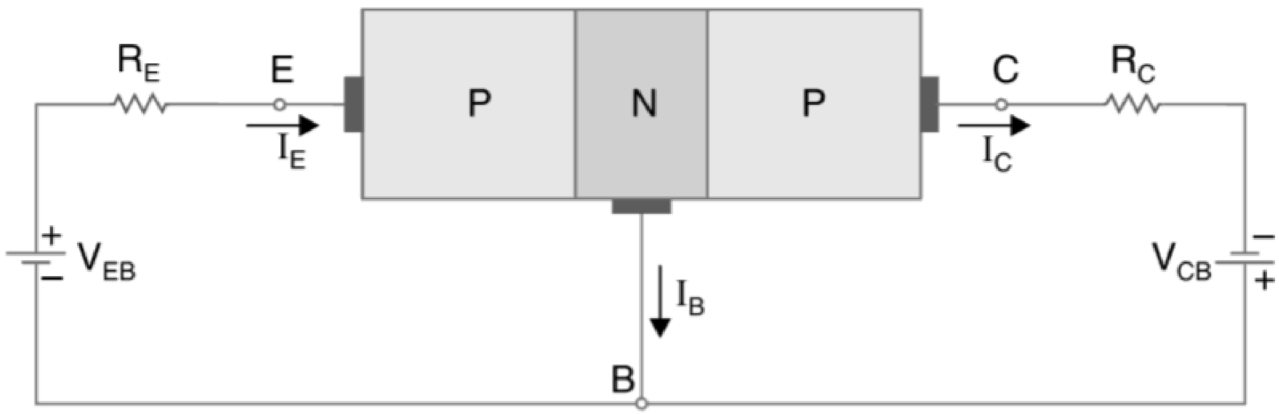
PNP Transistor

A PNP transistor consists of two P-type semiconductor regions separated by a thin N-type semiconductor region. The three regions of the transistor are called emitter, base, and collector. In a PNP transistor, the emitter-base junction is forward biased while the collector-base junction is reverse biased. Holes are the majority charge carriers in a PNP transistor, while electrons are minority carriers.

Working of PNP

- When the emitter-base junction is forward biased by the supply voltage V_{BE} , the heavily doped P-type emitter releases a large number of holes. These holes start moving from the emitter toward the base region due to the applied voltage. This movement of holes produces the emitter current I_E .
- The holes entering the base region recombine with a few free electrons present in the thin and lightly doped N-type base. This recombination produces a small base current I_B. Since the base region is very thin, only a small number of holes recombine while most holes continue moving toward the collector region.

Cont.



Cont.

- The collector-base junction is reverse biased by the supply voltage V_{CB} . Due to this reverse bias, the collector attracts the remaining holes coming from the base region. These holes enter the collector and produce collector current I_C .
- This collector current is also called injected current because it is produced due to the holes injected from the emitter region. Thus, most holes supplied by the emitter are collected by the collector.

Transistor Currents

1. **Emitter Current** $(\boldsymbol{I_E})$

Emitter current is the total current supplied by the emitter region. The emitter is heavily doped and injects majority charge carriers into the base region.

- In an NPN transistor, the emitter injects electrons.
- In a PNP transistor, the emitter injects holes.

Emitter current is the largest current in the transistor because it supplies charge carriers for transistor action.

Cont.

2. **Base Current** ($\boldsymbol{I_B}$)

When the charge carriers enter the base region, some of them recombine with opposite charge carriers present in the base.

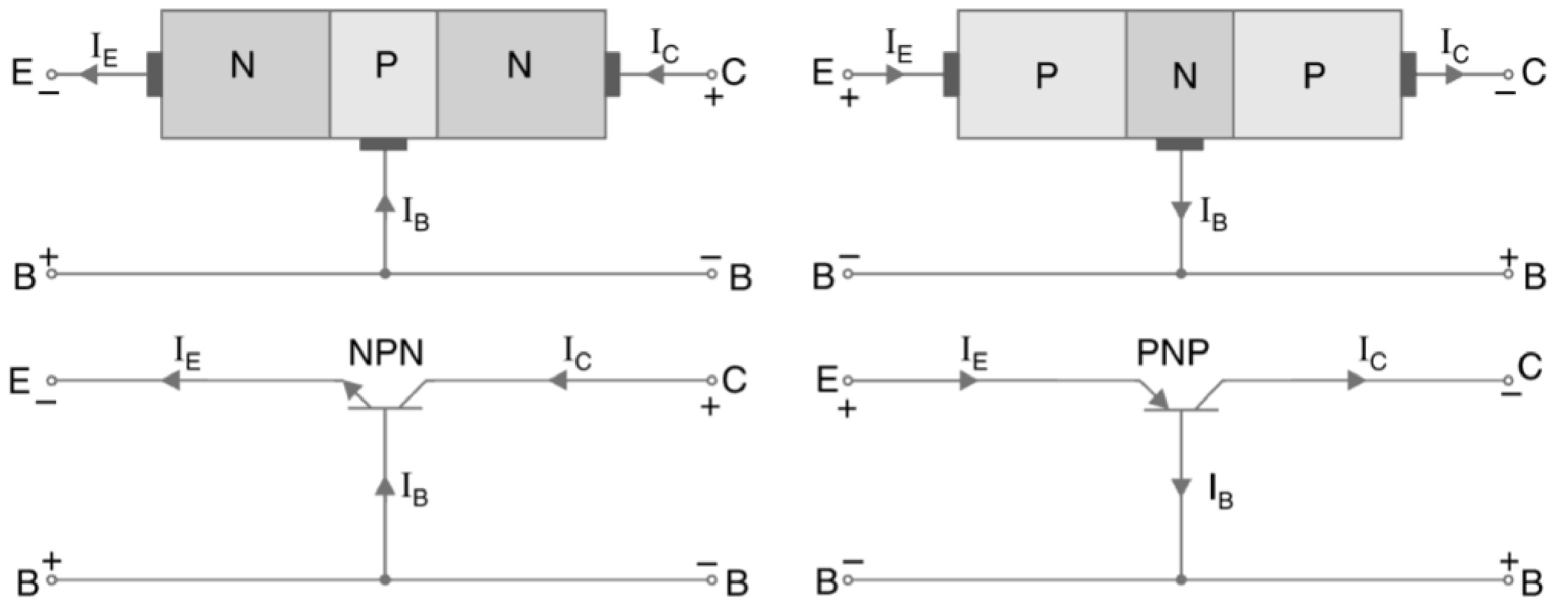
- In NPN transistor, electrons recombine with holes.
- In PNP transistor, holes recombine with electrons.

This recombination produces a small current called base current. The base region is very thin and lightly doped; therefore, only a small number of charge carriers recombine in the base. Hence, base current is very small compared to emitter and collector currents.

Cont.

3. **Collector Current** (I_C)

- Most of the charge carriers injected by the emitter do not recombine in the base region. These remaining carriers move toward the collector region. The collector-base junction is reverse biased, which attracts these charge carriers into the collector. This movement produces collector current.
- Collector current is the main current in a transistor and is also called injected current because it is produced due to the charge carriers injected from the emitter region.



(a) NPN transistor currents.

(b) PNP transistor currents.

Current Equation of Transistor

The emitter current is equal to the sum of base current and collector current:

$$I_E = I_B + I_C$$

Since the base current is very small:

$$I_C \approx I_E$$

Current Gain (Beta)

The current gain of a transistor is represented by beta (β) and is given by:

$$\beta = \frac{I_C}{I_B}$$

Beta shows how effectively a transistor amplifies current.

It shows how a small base current controls a much larger collector current.

- Example: If β=100

then the collector current is 100 times greater than the base current.

Thus, a small input current at the base controls a large output current at the collector.

This property makes the transistor useful as an amplifier and as an electronic switch.

Current Gain (Alpha)

In a transistor, **alpha (∝)** is the **current gain in the common-base configuration**.

$$\propto = \frac{I_c}{I_E}$$

It tells how much of the emitter current reaches the collector. Since some current is lost in the base, I_c is slightly less than I_E.

- For a normal transistor:
- $(\propto) \approx$ 0.95 to 0.990 (always less than 1)

Relation Between Current Gain α and β

We know that emitter current (I_E) of a transistor is the sum of its base current (I_B) and collector current (I_C). *i.e.*,

$$I_E = I_B + I_C$$

Dividing the above equation on both sides by I_C,

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + 1$$

Since $I_C/I_E = \alpha$ and $I_C/I_B = \beta$ therefore

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1 = \frac{1+\beta}{\beta}$$

$$\therefore \qquad \alpha = \frac{\beta}{\beta + 1}$$

The above expression may be written as

$$\alpha(\beta + 1) = \beta$$

$$\alpha \cdot \beta + \alpha = \beta$$

$$\alpha = \beta - \alpha \cdot \beta = \beta(1 - \alpha)$$

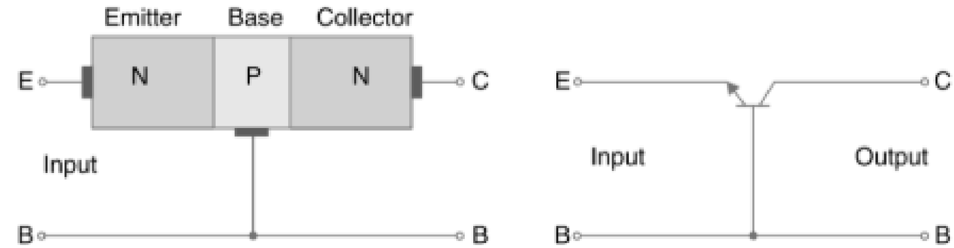
$$\therefore \qquad \beta = \frac{\alpha}{1 - \alpha}$$

BJT Circuit Configurations

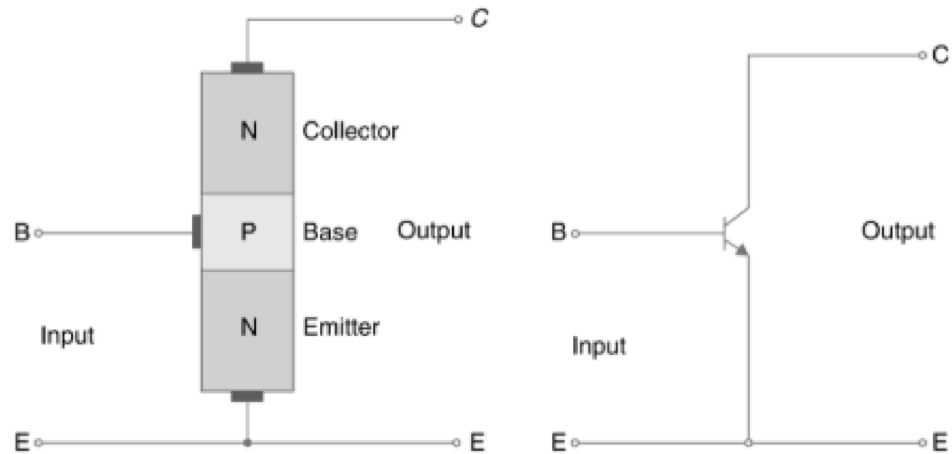
We know that a transistor has three terminals or leads namely emitter (E), base (B) and collector (C). However, when a transistor is connected in a circuit, we require four terminals *i.e.*, two terminals for input and two for output. This difficulty is overcome by using one of the three terminals as a common terminal to the input and output terminals. Depending upon the terminals, which are used as a common terminals, the transistors can be connected in the following three different connections or configurations:

1. ***Common-base (CB) connection.*** In this configuration, the transistor is connected with the base as a common terminal as shown in Fig. ——— (*a*). The input is applied between the emitter and base terminals. The output is taken between the collector and base terminals. This type of configuration is used to explain the operation of NPN and PNP transistors.

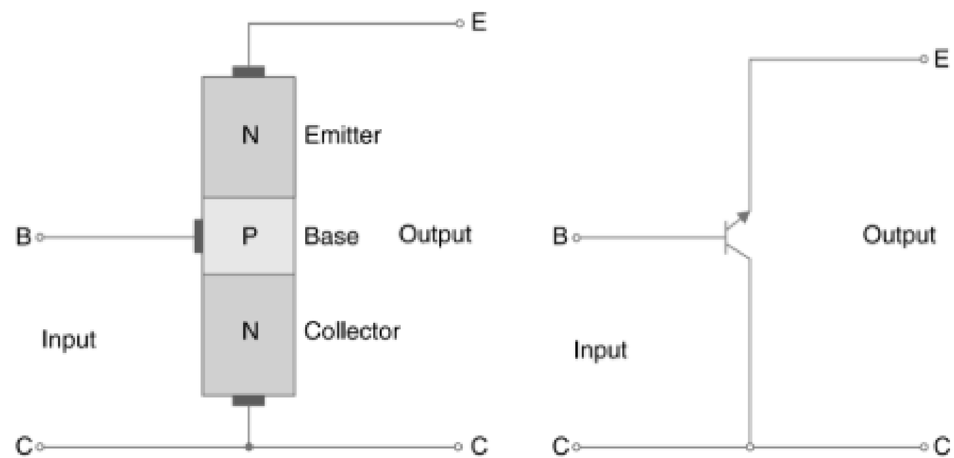
2. ***Common-emitter (CE) connection.*** In this configuration, the transistor is connected with the emitter as a common terminal as shown in Fig. ——— (*b*). The input is applied between the base and collector terminals. The output is taken between collector and emitter terminals. This is one of the most commonly used connections of a transistor.
3. ***Common-collector (CC) connection.*** In this configuration, the transistor is connected with collector as a common terminal as shown in Fig. ——— (*c*). The input is applied between the base and collector terminals. The output is taken between emitter and collector terminals.



(*a*) Common-base connection.



(*b*) Common-emitter connection.



(*c*) Common-collector connection.

NPN vs PNP Transistor

NPN

- It is made of a combination of two N-layers and one P-layer.
- The majority charge carriers are electrons. The minority charge carriers are holes.
- In the NPN symbol, the emitter arrow is pointing outward.

PNP

- It is made up of two P-layers and one N-layer.
- The majority charge carriers are holes. The minority charge carriers are electrons.
- In the PNP symbol, the emitter arrow is pointing inward.

Cont.

NPN

- It starts conduction once electrons enter the base region.
- The current flows from collector towards emitter.
- The base current goes into the base through the emitter.
- It has a small switching time thus having a high switching speed

PNP

- It starts conduction once the holes enter the base region.
- The current flows from emitter to collector.
- The base current goes from the emitter into the base.
- It has a high switching time thus having a low switching speed

Cont.

NPN

- It switches ON by applying a positive voltage to the base.
- It switches OFF when the base voltage becomes low or zero.
- Widely used in low-side switching and digital control circuits such as microcontrollers, relays, and LED drivers.

PNP

- It switches ON by applying a negative or lower voltage to the base.
- It switches OFF when the base voltage becomes positive relative to the emitter
- Widely used in high-side switching and complementary stages in amplifiers and power control circuits.